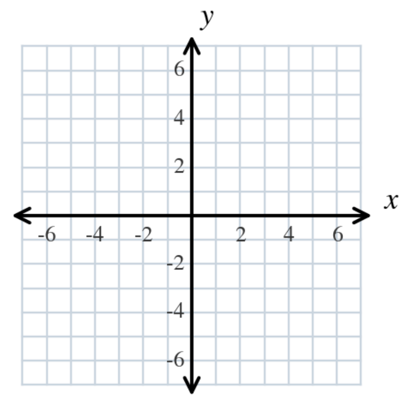
SC Honors Geometry - Nappier
4.1 – Translations • Homework

Name: ___________________________________



Find the image of each point after the given translation.

1. Translate $A(3, -2)$ along the vector $\langle -5, 4 \rangle$.

2. Translate $B(-1, -6)$ along the vector $\langle 2, 7 \rangle$.

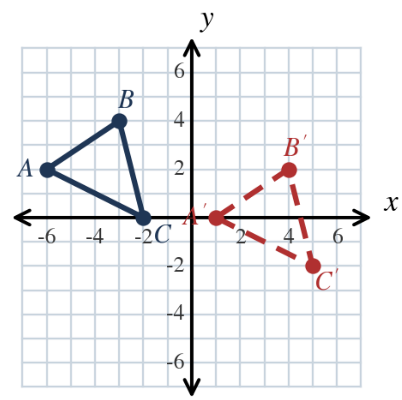
Find the component form of the vector.

3. The vector translates $C(-4, 1)$ to $C'(2, -3)$.

4. The vector translates $D(5, 0)$ to $D'(-1, 6)$.

Use the coordinate plane at the right.

5. Graph $\triangle FGH$ with $F(-4, 1)$, $G(-1, 3)$, and $H(-2, -2)$ and its image after the translation $(x, y) \rightarrow (x + 6, y - 4)$. List the image coordinates.



Use the translation $(x, y) \rightarrow (x - 3, y + 7)$.

6. What is the image of $J(4, -5)$?

7. What is the preimage of $K'(-2, 3)$?

Use the second diagram at the right.

8. Write a rule for the translation of $\triangle ABC$ to $\triangle A'B'C'$.

Answer each question.

9. $\triangle LMN$ is translated along $\langle -4, -1 \rangle$ and then along $\langle 6, 5 \rangle$. Write ONE rule for the composition.

10. A translation maps $P(-2, 1)$ to $P'(4, 9)$. Find PP'. Then explain why every other point of the figure moves that same distance.